Chemistry - Graduate Level Assessment 10 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Calculate the normality of a solution containing 2.45 g of sulfuric acid in 2.00 liters of solution. (MW of H_2 SO_4 = 98.1.)
   (a) 0.0125 equiv/liter
   (b) 0.060 equiv/liter
   (c) 0.050 equiv/liter
   (d) 0.100 equiv/liter
   (e) 0.125 equiv/liter

2. Calculate the equilibrium polarization of 13 C nuclei in a 20.0 T magnetic field at 300 K.
   (a) 7.39 x $10^{-}5$
   (a) 9.12 x $10^{-}5$
   (a) 3.43 x $10^{-}5$
   (a) 6.28 x $10^{-}5$
   (a) 1.71 x $10^{-}5$

3. If we locate an electron to within 20pm, then what is the uncertainty in its speed? $
   (a) $8.110^{6}\ \mathrm{m \cdot s^{-1}}$
   (b) $9.410^{7}\ \mathrm{m \cdot s^{-1}}$
   (c) $6.210^{7}\ \mathrm{m \cdot s^{-1}}$
   (d) $1.110^{8}\ \mathrm{m \cdot s^{-1}}$
   (e) $3.710^{7}\ \mathrm{m \cdot s^{-1}}$

4. Compare the bond order of He_2 and He_2^{+}.
   5. Bond order of He_2 is 0 and He_$2^{+} is 0.5$
   5. Bond order of He_2 is 0.5 and He_$2^{+} is 0$
   5. Bond order of He_2 and He_$2^{+} is 0.5$
   5. Bond order of He_2 is 2 and He_$2^{+} is 1.5$
   5. Bond order of He_2 is 1 and He_$2^{+} is 0$

A solution of sulfurous acid, H_2SO_3, is present in an aqueous solution. Which of the following represents the concentrations of three different ions in solution?

H_2SO_3 > [HSO 3-] = [SO 32-]

HSO 3- > [SO 32-] > [H_2SO_3]

SO 32- > [H_2SO_3] > [HSO 3-]

HSO 3- = [SO 32-] > [H_2SO_3]

SO 32- = [HSO 3-] > [H_2SO_3]

True / False (10 marks)

Answer True or False

6. True or False: The volume of a cube with edges measuring 150.0 mm is 450 cubic centimeters.

7. True or False: The molecule NO 2- exhibits sp 2 hybridization around its central nitrogen atom.

8. True or False: The recoil energy of the atom is 400 eV, which is the expected value for such emissions in similar scenarios.

9. True or False: The reduction of NO(g) typically results in the formation of nitrogen dioxide NO 2(g) as a product.

10. True or False: Electrons can occupy the same orbital if they possess identical quantum numbers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between temperature and the standard molar enthalpy of sulfur dioxide gas, calculating the increase in enthalpy as the temperature rises from 298.15 K to 1500 K.

12. Discuss the conditions under which precipitates of ZnS and CuS would form when H_2S is bubbled into a solution containing Zn^2+ and Cu^2+ ions at 0.02 M each, and analyze the implications of their solubility products.

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